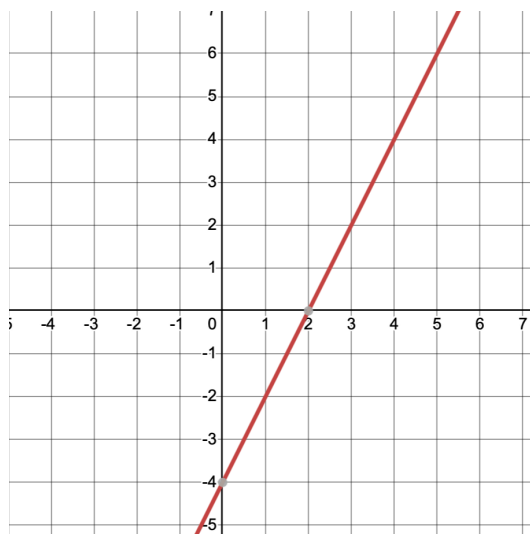
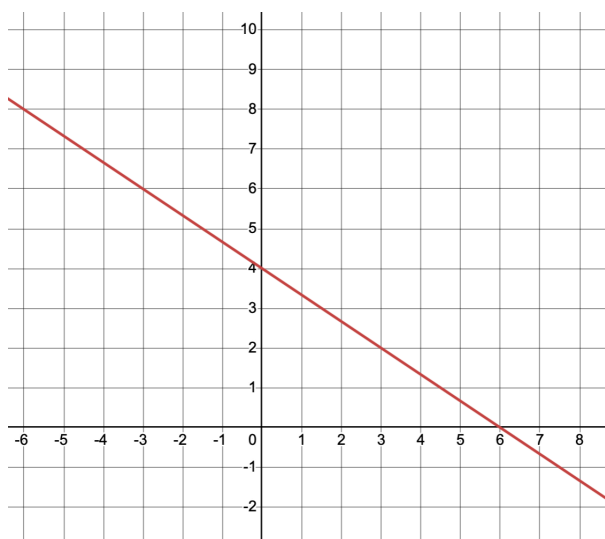


(review) Test 2 review

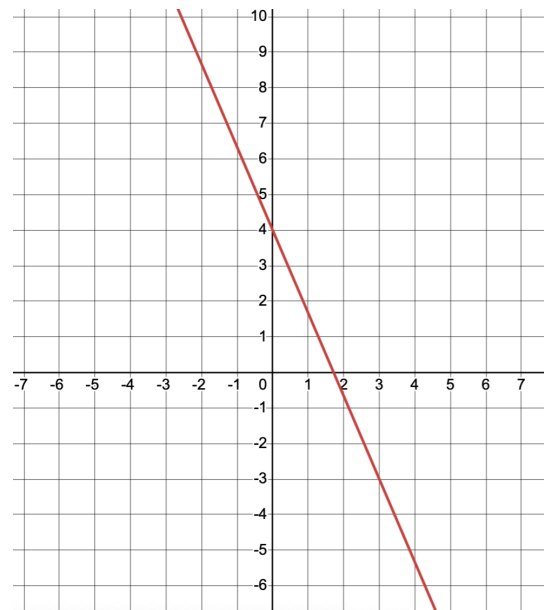
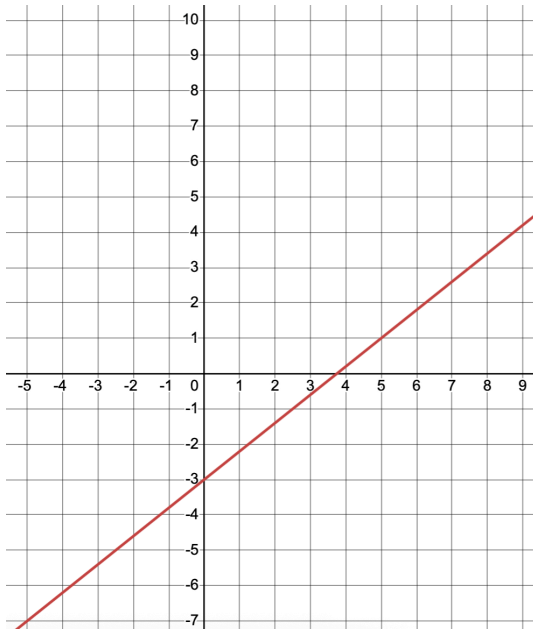
Lines

Equation for line

- Given 2 points
 - Find the **slope intercept form** of the line that goes through points $(-2, -3)$ and $(3, 12)$
 - Find the slope intercept form of the line that goes through points $(-3, 3)$ and $(1, -5)$
 - Find the slope intercept form of the line that goes through points $(-1, 3)$ and $(1, 4)$
 - Find the slope intercept form of the line that goes through points $(-3, 2)$ and $(5, 4)$
 - Find the slope intercept form of the line that goes through points $(-1, -5)$ and $(2, 3)$
- given 1 point and the slope
 - Find the slope intercept form of the line that goes through the point $(-3, 1)$ with a slope of $m = 2$
- given the y-intercept and the slope
- Parallel, perpendicular
 - Write the equation for the line parallel to $y = 2x + 3$ and passes through point $(4, 10)$
 - Write the equation for the line parallel to $y = 4x - 4$ and passes through point $(3, 15)$
 - Write the equation for the line parallel to $y = -3x - 3$ and passes through $(3, -14)$
- slope and equation for vertical and horizontal lines
- Given a graph, identify the slope and y-intercept to write the equation of the linear function.



Given a graph, identify the slope and y-intercept to write the equation of the linear function.



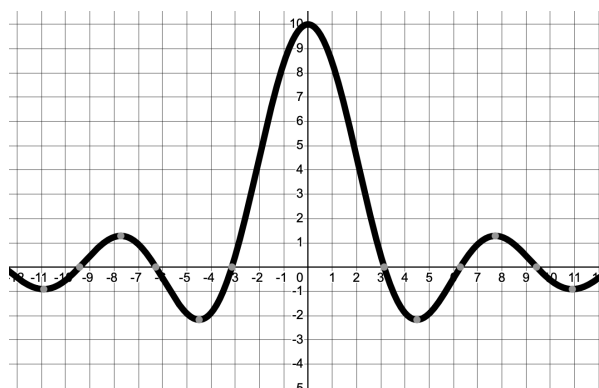
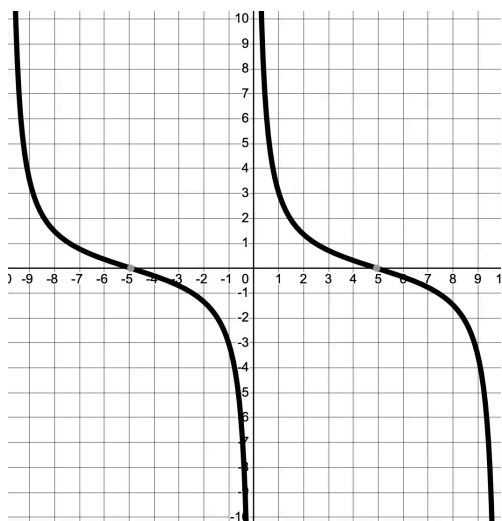
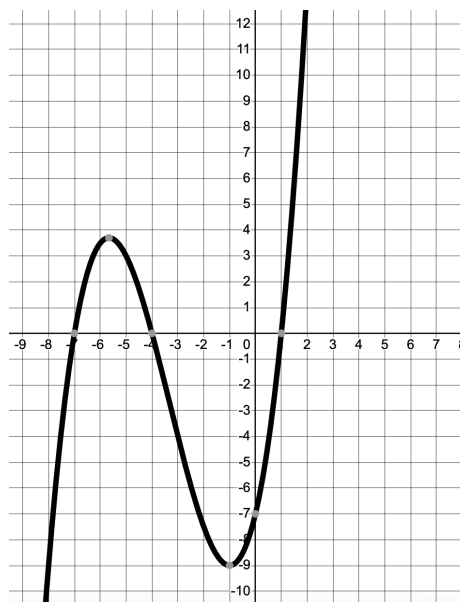
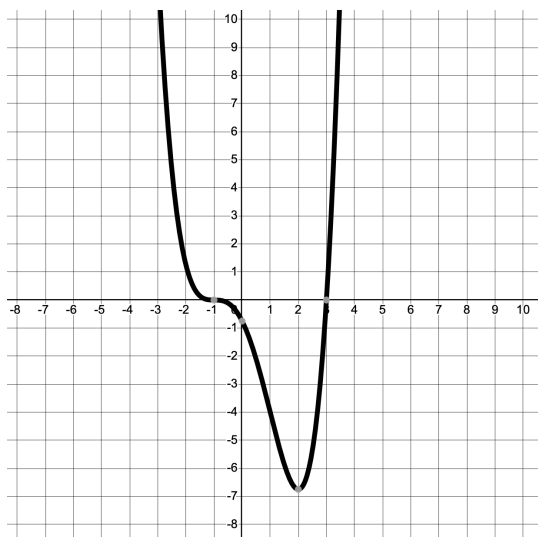
Linear Functions and word problems

- Diego has a pet shark that cost \$15. Every month, he spends \$20 to feed it. Write a linear function for the cost of Diego's shark
 - What is the base cost and what is the rate (include units)?
 - How much will he spend in 1 year
 - How many months will it take for him to spend \$500?
- Kaley spends an average of \$40 a month at Taco Bell. Write a linear function for how much Kaley spends at Taco Bell
 - What is the base cost and what is the rate (include units)?
 - How much will it cost after 6 months?
 - How long will it take for her to spend \$1000 (the cost of rent)?

Behavior of graphs

For each of the graphs, you should be able to:

1. Domain and Range
2. Find the extrema (minimum/maximums)
 - Determine if they are global or local min/max
 - Identify **where** the extrema occurs (x -value)
 - Determine the **value** of the extrema (y -value)
3. Identify the x -interval where the function is increasing/decreasing
4. Report the x -intercepts, y -intercepts
5. Over a given x -interval, calculate the average rate of change



Quadratics

Factor

1. $y^2 - 11y + 24$

2. $x^2 - 2x - 35$

3. $x^2 + 8x + 7$

4. $x^2 - 2x - 48$

5. $2x^2 - x - 15$

6. $2x^3 - 128x$

1. $2x^2 - 9x + 10$

2. $x^2 - 24x + 144$

3. $x^2 - 16x + 64$

4. $x^2 + 22x + 121$

5. $4x^2 - 64$

6. $49x^2 - 4$

Use factoring to solve for x-int

1. $f(x) = x^2 + 11x - 26$

2. $f(x) = x^2 + 5x - 66$

3. $f(x) = x^2 - 12x$

4. $f(x) = x^2 + 2x - 63$

5. $f(x) = 3x^2 + 10x - 8$

6. $f(x) = 5x^2 - 9x + 4$

7. $f(x) = 5x^2 - 14x + 8$

8. $f(x) = 2x^2 + 13x + 15$

Completing the square to get vertex coordinate point

1. $y = x^2 + 16x + 16$

2. $y = x^2 + 8x + 7$

**Complete the square to write the vertex form. Then, report the vertex point.
Put the function in factored form. Then, report the x-intercepts.
Sketch the quadratic function.**

1. $f(x) = x^2 + 10x + 24$

2. $f(x) = x^2 - 8x + 12$

3. $f(x) = x^2 + 8x + 15$

4. $f(x) = x^2 - 6x + 8$

5. $y = x^2 - 4x - 12$

6. $y = x^2 - 6x - 16$

7. $y = x^2 - 2x - 35$

8. $y = x^2 - 4x - 32$